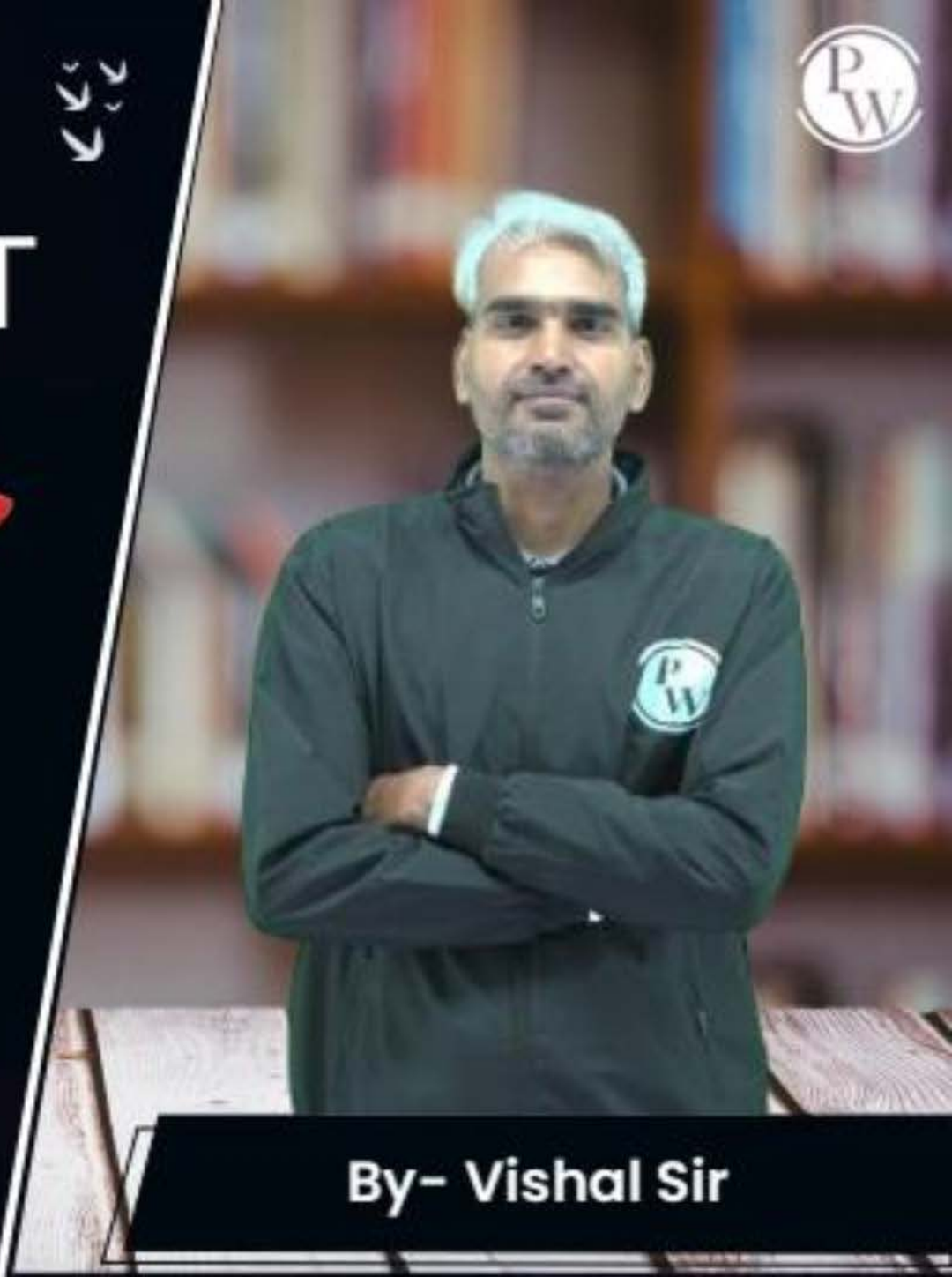
Computer Science & IT



Database Management System

Query Languages

Lecture No. 03



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Derived relational algebra operations
- **Topic** Join operations
- **Topic** Division operation

Slide

Topics to be Covered

Topic

Practice questions



Slide

Consider the following relational tables:

Supplier (<u>Sid</u>, Sname, Rating)

Parts (<u>Pid</u>, Pname, Color)

Catalog (<u>Sid</u>, <u>Pid</u>, Cost)

Supplier (S)

Sid	Sname	Rating
S_1	A	3
S_2	B	5
S_3	A	7
S_4	A	0

Catalog

Sid	Pid	Cost
S_1	P_1	20
S_1	P_2	30
S_2	P_2	30
S_3	P_3	10

Parts (P)

Pid	Pname	Color
P_1	XYZ	Red
P_2	PQR	Green
P_3	LMN	Red

#Q. Retrieve Sid of the suppliers whose rating is more than 5.

$$\pi_{Sid}\left(\sigma_{Rating>5}\left(Supplier\right)\right)$$

#Q. Retrieve Sid of the suppliers who have supplied some parts.

π_{sid} (Supplier) Wrong.

π_{sid} (Catalog) ✓

#Q. Retrieve Sid of the suppliers who have supplied some Red color parts.

100 x 200 = 20000

(i) $\pi_{Catalog.Sid}\left(\sigma_{Catalog.Pid=Parts.Pid \wedge Parts.Color='RED'}(Catalog \times Parts)\right)$

Only 20 are w.r.t Red Color Parts

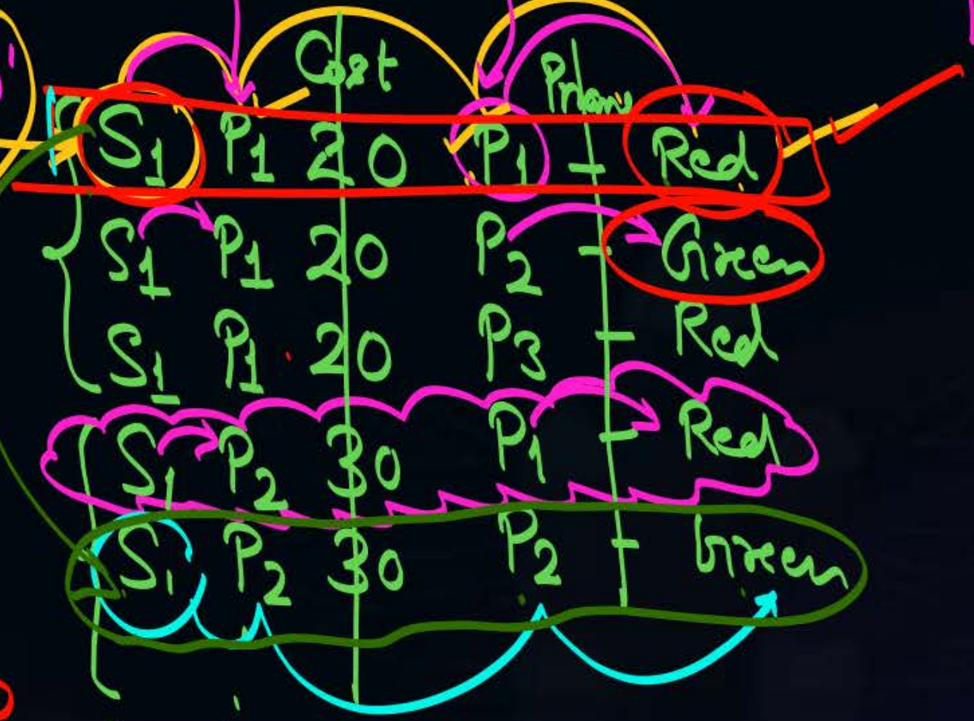
(ii) $\pi_{Sid}\left(\sigma_{Color='RED'}(Catalog \bowtie Parts)\right)$

100 x 200 = 20000

(iii) $\pi_{Sid}\left(Catalog \bowtie \left(\sigma_{Color='RED'}(Parts)\right)\right)$

100 x 20 = 2000

→ Most efficient query among above three.



#Q. Retrieve Sid of the suppliers who have supplied some Red or some Green color parts.

(i) $\pi_{Catalog.Sid}\left(\sigma_{(Catalog.pid = Parts.Pid}(Catalog \times Parts) \wedge (Parts.Color = 'RED' \vee Parts.Color = 'Green'))\right)$

(ii) $\pi_{Sid}\left(\sigma_{(Color = 'RED' \vee Color = 'Green')}(Catalog \bowtie Parts)\right)$

(iii) $\pi_{Sid}\left(Catalog \bowtie \left(\sigma_{(Color = 'RED' \vee Color = 'Green')}(Parts)\right)\right)$

#Q. Retrieve Sid of the suppliers who have supplied some Red or some Green color parts.

(iv)

$$\pi_{C_1.Sid}\Big[\sigma_{(C_1.Sid = C_2.Sid) \wedge ((C_1.Pid = P_1.Pid \wedge P_1.Color='RED') \vee (C_2.Pid = P_2.Pid \wedge P_2.Color='GREEN'))}\Big[\underbrace{(\rho_{C_1}(Catalog) \times \rho_{P_1}(Parts))}_{(C_1 \times P_1)} \times \underbrace{(\rho_{C_2}(Catalog) \times \rho_{P_2}(Parts))}_{(C_2 \times P_2)}\Big]\Big]$$

used to select the Suppliers w.r.t 'RED' Color

Used to select the Supplier w.r.t 'Green' Color

#Q. Retrieve Sid of the suppliers who have supplied some Red and some Green color parts.

(1)

$$\left[\pi_{Sid}\left(C_1 \bowtie \sigma_{Color='RED'}(P_1)\right) \;\cup\; \pi_{Sid}\left(C_2 \bowtie \sigma_{Color='GREEN'}(P_2)\right)\right]$$

S_1
S_3

S_1
S_2

S_1

#Q. Retrieve Sid of the suppliers who have supplied some Red and some Green color parts.

(i) $\pi_{Catalog.Sid}\left(\sigma_{(Catalog.Pid = Parts.Pid}(Catalog \times Parts)\right)$

$(Parts.Color = 'RED' \wedge Parts.Color = 'Green'))$

(ii) $\pi_{Sid}\left(\sigma_{(Color = 'RED' \wedge Color = 'Green')}(Catalog \bowtie Parts)\right)$

(iii) $\pi_{Sid}\left(Catalog \bowtie \left(\sigma_{(Color = 'RED' \wedge Color = Green)}(Parts)\right)\right)$

All are wrong

#Q. Retrieve Sid of the suppliers who have supplied some Red and some Green color parts.

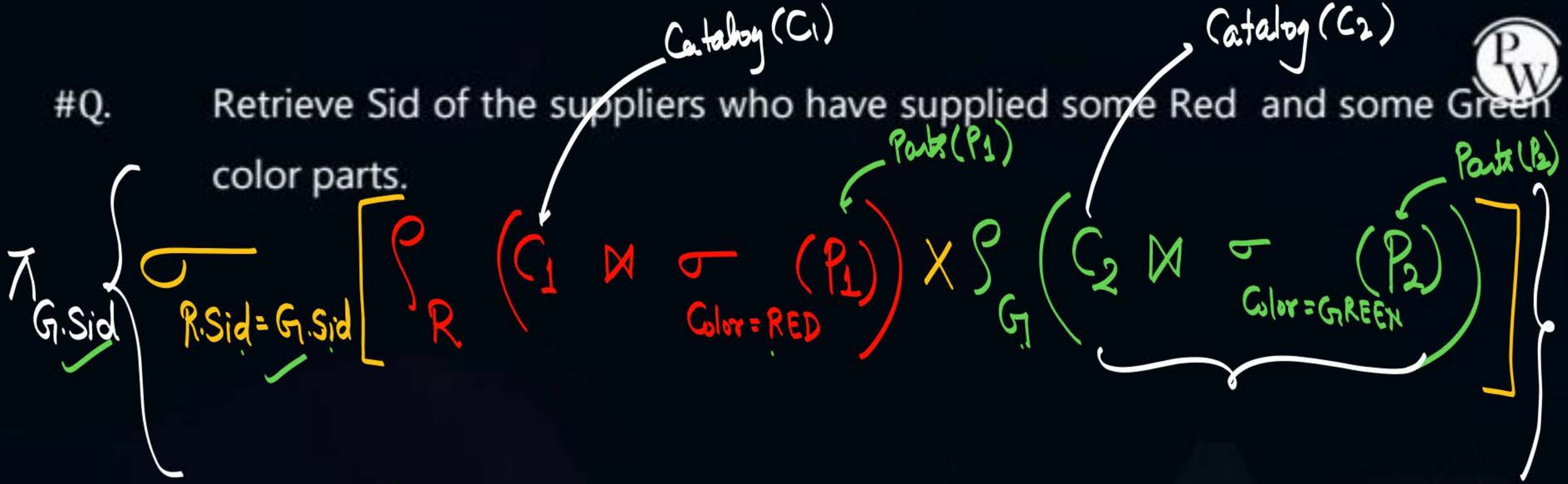
$$\pi_{C_1.Sid}\Big[\sigma_{C_1.Sid = C_2.Sid}\Big[\big(\rho_{C_1}(Catalog) \times \rho_{P_1}(Parts)\big) \times \big(\rho_{C_2}(Catalog) \times \rho_{P_2}(Parts)\big)\Big]\Big]$$

$$\sigma_{(C_1.Pid = P_1.Pid \wedge P_1.Color = 'RED') \wedge (C_2.Pid = P_2.Pid \wedge P_2.Color = 'GREEN')}$$

$(C_1 \times P_1)$ used to select the suppliers w.r.t 'RED' Color

$(C_2 \times P_2)$ Used to select the supplier w.r.t 'Green' Color

#Q. Retrieve Sid of the suppliers who have supplied some Red and some Green color parts.



#Q. Retrieve Sid of the suppliers who have supplied some Red and some Green color parts.

$$\left[\pi_{Sid}\left(C_1 \bowtie \sigma_{Color='RED'}(P_1)\right)\right] \wedge \left[\pi_{Sid}\left(C_2 \bowtie \sigma_{Color='GREEN'}(P_2)\right)\right]$$

S_1
S_3

S_1
S_2

S_1

#Q. Retrieve Sid of the suppliers who have supplied all Red color parts.

Division

$$\pi_{sid,pid}(Catalog) \div \pi_{pid}\left(\sigma_{Color='RED'}(Parts)\right)$$

$$= \left[\pi_{sid}(Catalog) - \pi_{sid}\left[\left(\pi_{sid}(Catalog) \times \pi_{pid}\left(\sigma_{Color='RED'}(Parts)\right)\right) - \pi_{sid,pid}(Catalog)\right]\right]$$

- $\pi_{sid}(Catalog)$: Suppliers who have supplied all red Color Parts.
- π_{sid}: Suppliers who did not supply all red Color Parts
- $\pi_{sid}(Catalog)$: Suppliers who have supplied at least One Part
- $\pi_{pid}\left(\sigma_{Color='RED'}(Parts)\right)$: Pids of all Red Color Parts
- $\times$: Every Supplier in Catalog table Supplied all Red Color Parts
- $\pi_{sid,pid}(Catalog)$

#Q. Retrieve Sid of the suppliers who have supplied all Red color parts.

Division

$$\pi_{Sid,Pid}(Catalog) \div \pi_{Pid}(\sigma_{Color='RED'}(Parts))$$

$$= \left[\pi_{Sid}(Catalog) - \pi_{Sid}\left[\left(\pi_{Sid}(Catalog) \times \pi_{Pid}(\sigma_{Color='RED'}(Parts))\right) - \pi_{Sid,Pid}(Catalog)\right]\right]$$

Sid
S_1
S_2
S_3

Pid
P_1
P_3

Sid	Pid
S_1	P_3
S_2	P_1
S_2	P_3
S_3	P_1

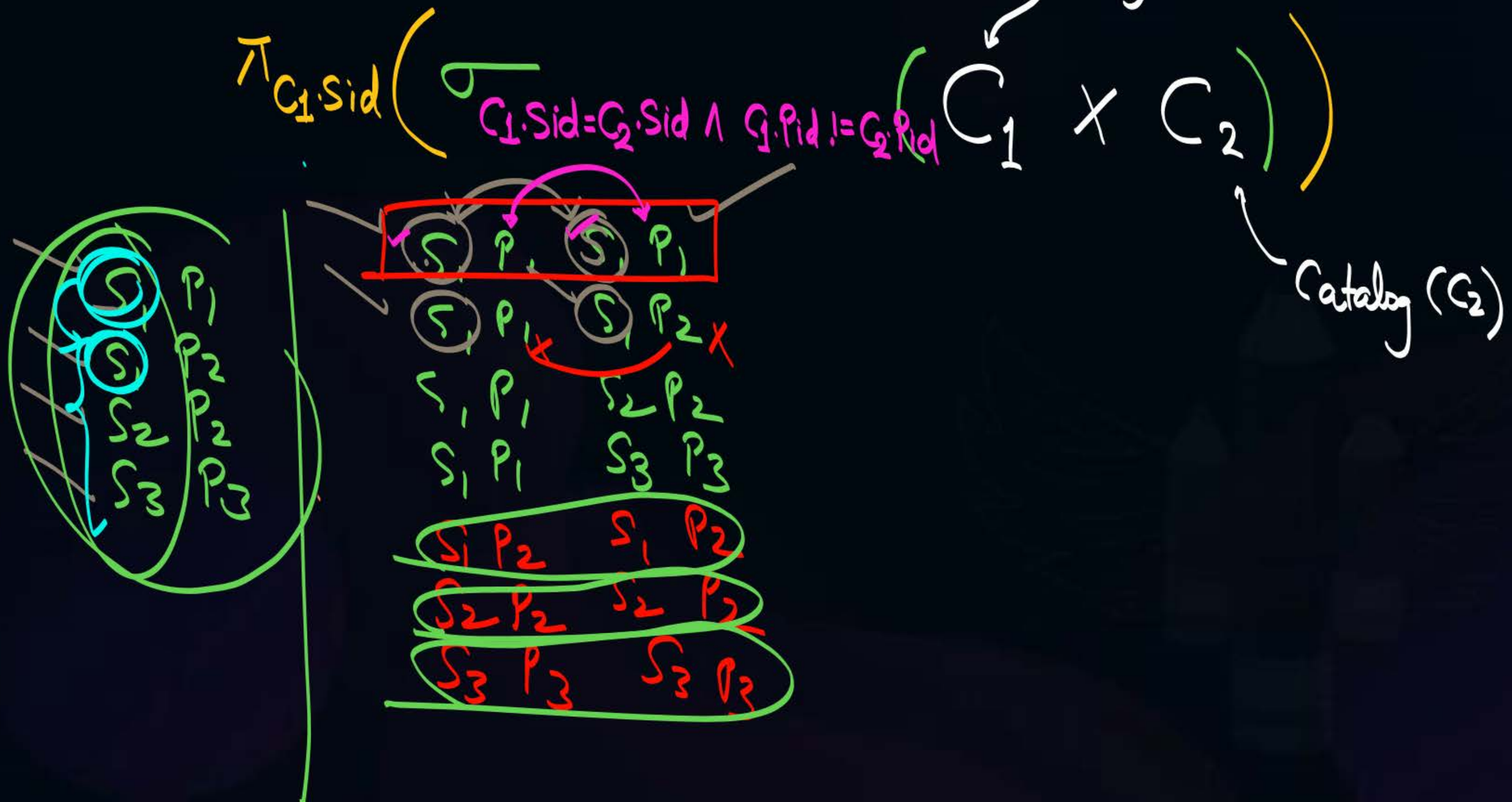
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#Q. Retrieve Sid of the suppliers who have supplied at least two parts.

$$\pi_{C_1.Sid}\left(\sigma_{C_1.Sid=C_2.Sid \wedge C_1.Pid != C_2.Pid}\left(C_1 \times C_2\right)\right)$$

Catalog (C_1)

Catalog (C_2)



#Q. Retrieve Sid of the suppliers who have supplied exactly one part.

π_{sid} (Catalog) — { Suppliers who have Supplied at least two parts }

Suppliers who have supplied at least one part

i.e. {1, 2, 3, ---}

{2, 3, 4, ---}

Suppliers who have Supplied exactly one part

#Q. Retrieve Sid of the suppliers who have supplied at most one part.

0 or 1

π_{Sid} (Supplies) $-$ { Supplies who have supplied at least two parts }

↓

Supplies who have supplied any number of parts

{ i.e. 0, 1, 2, 3, ---. }

{0, 1}

i.e. at most '1'.

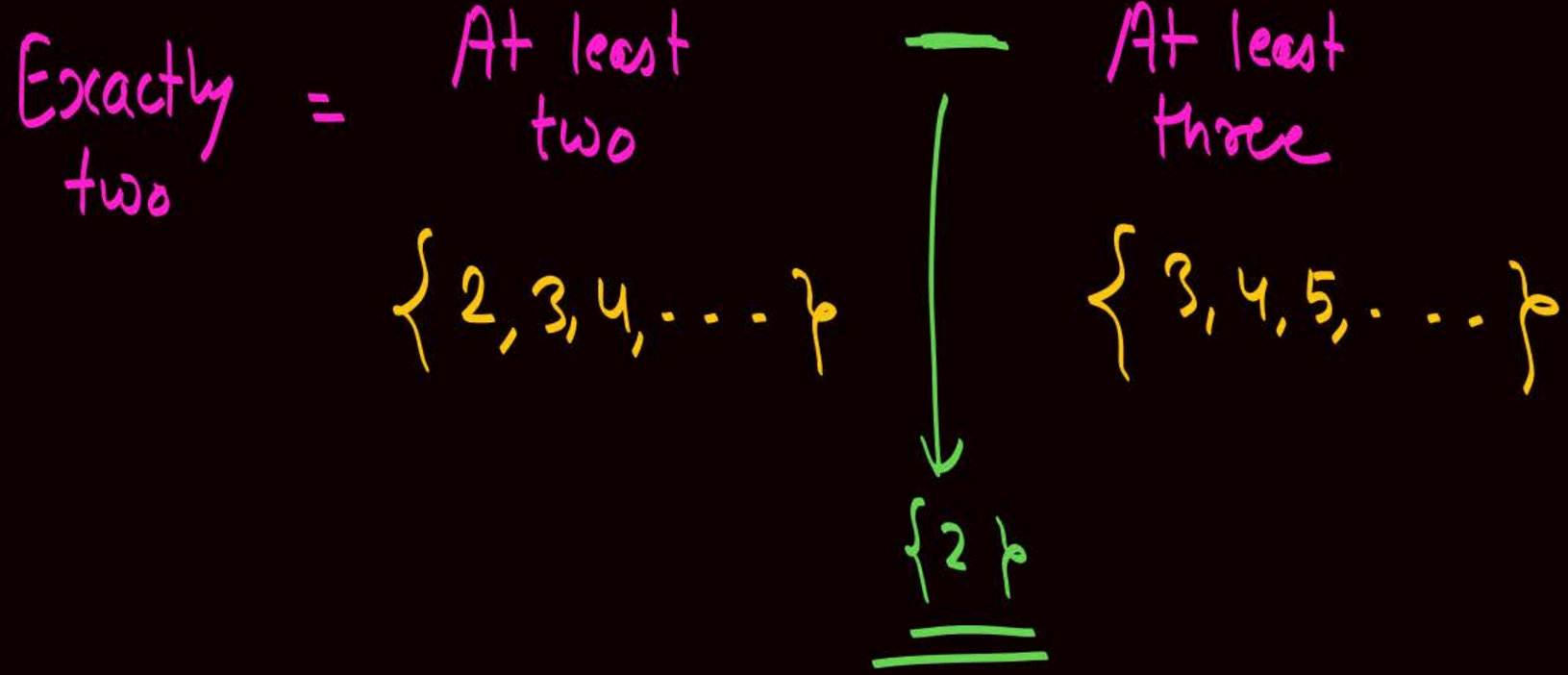
{ i.e., 2, 3, 4, ---. }

#Q. Retrieve Sid of the suppliers who have supplied at least three parts.

we don't need to Compare $C_1.Sid = C_3.Sid$

$$\Pi_{C_1.Sid}\left[\sigma\left(\neg((C_1.Sid=C_2.Sid)\wedge(C_2.Sid=C_3.Sid)) \wedge (C_1.Pid!=C_2.Pid \wedge C_2.Pid!=C_3.Pid \wedge C_1.Pid!=C_3.Pid)\right)(C_1 \times C_2 \times C_3)\right]$$

Q: Retrieve Sids of the suppliers who supplied exactly two parts?



H.W:

#Q. Retrieve Sid of the suppliers who have supplied most expensive parts.



2 mins Summary

Topic Relational algebra

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THANK - YOU